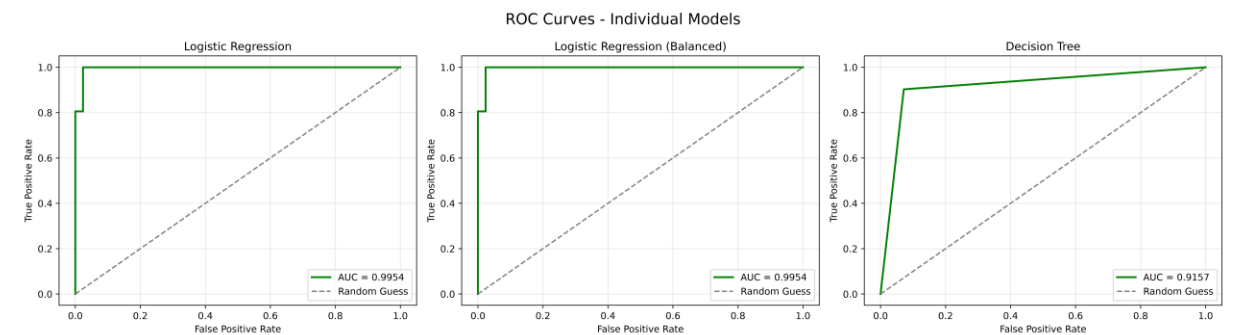


Classification Models, Evaluation Metrics & Handling Imbalanced Data

AI & Machine Learning — Task 4

Why These Metrics Were Chosen

- Dataset: Breast Cancer (569 samples) — Malignant vs Benign
- Accuracy alone is misleading: it treats a missed cancer case the same as a false alarm
- Recall prioritized — a False Negative (missed Malignant) is the costliest error in diagnosis
- F1-Score used as the primary comparison metric — balances Precision & Recall, doesn't get inflated by the majority class
- AUC used to judge overall class separation independent of threshold



Handling Class Imbalance

- Class split: Benign 62.7% / Malignant 37.3% — mild imbalance
- Technique applied: `class_weight="balanced"` in Logistic Regression
- Result:

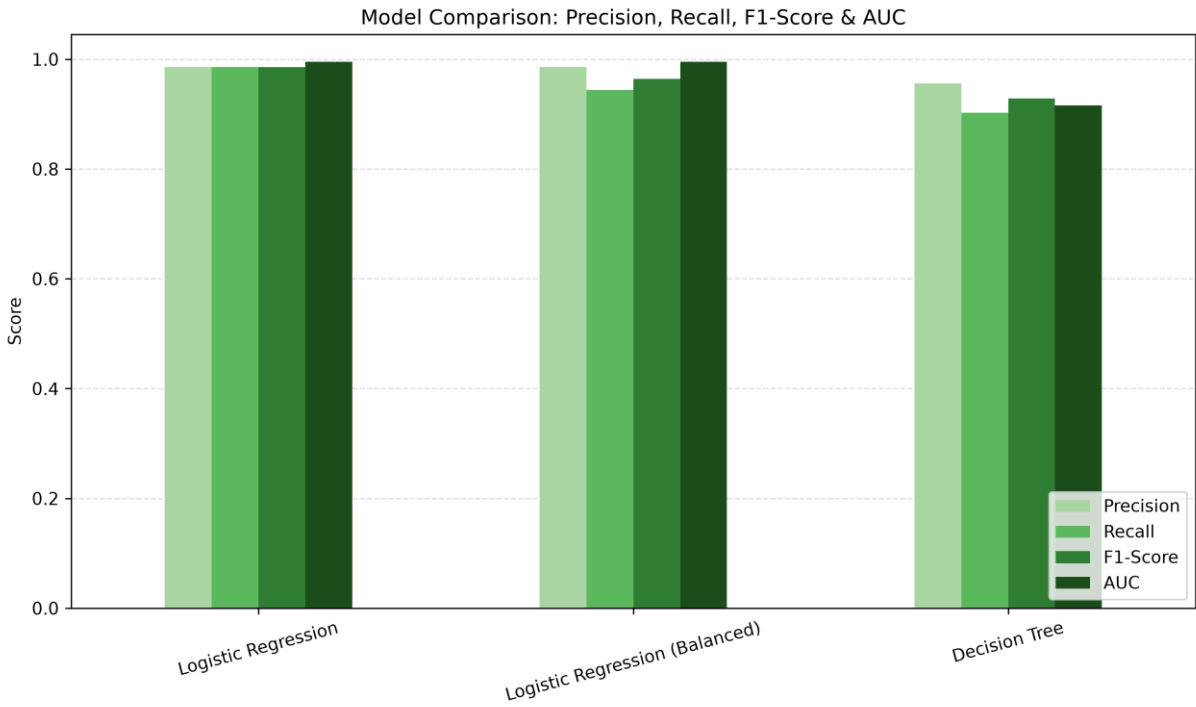
Model	Precision	Recall	F1
Baseline	0.9861	0.9861	0.9861
Balanced	0.9855	0.9444	0.9645

- *On this dataset, balancing reduced precision without improving recall — the imbalance wasn't severe enough to need correction*

Model Comparison

Model	Accuracy	F1-Score	AUC
Logistic Regression	0.9825	0.9861	0.9954
Logistic Regression (Balanced)	0.9561	0.9645	0.9954
Decision Tree	0.9123	0.9286	0.9157

- Decision Tree overfit: Train Acc 100% vs Test Acc 91.2% (gap 0.088)
- Logistic Regression generalized best (gap 0.007)



Final Model Decision

Selected Model: Logistic Regression (baseline)

- Highest Accuracy, F1-Score, and Recall of all models tested
- Most stable (smallest train-test gap) and most interpretable
- Class-weighting was tested for imbalance but didn't outperform the baseline here
- Conclusion: simplest model won because it already generalized well — added complexity wasn't needed